Class - 13

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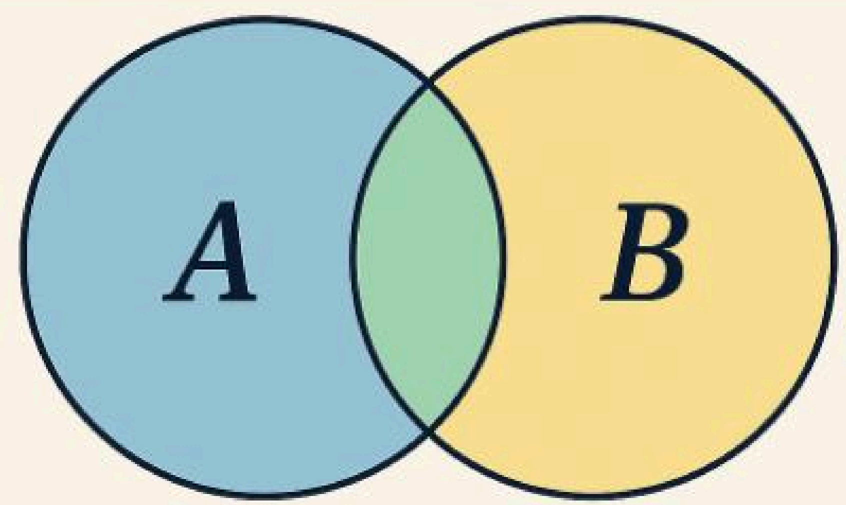
Bayes Theorem

Probability

BAYES THEOREM

Bayes' theorem is a rule in probability that helps us **update the probability of an event based on new information (evidence).**

Bayes theorem basically **update or change the probability** when we know the answer of that event.



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BAYES THEOREM

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- $P(A|B)$ = probability of event A happening **given** B is true (posterior).
- $P(B|A)$ = probability of seeing B if A is true (likelihood).
- $P(A)$ = initial probability of A (prior).
- $P(B)$ = probability of B happening (evidence/normalizer).

BAYES THEOREM

Example:

- A school has **60% boys** and **40% girls**.
- 30% of boys wear glasses.
- 50% of girls wear glasses.

$$P(Girl|Glasses) = \frac{P(Glasses|Girl) \cdot P(Girl)}{P(Glasses)}$$

- $P(Girl)=0.4$
- $P(Boy)=0.6$
- $P(Glasses|Girl)=0.5$
- $P(Glasses|Boy)=0.3$

$$P(Girl|Glasses) = \frac{0.5 \times 0.4}{0.38} = \frac{0.20}{0.38} \approx 0.526$$

BAYES THEOREM

To basically bayes theorem hamare intuition kitna sahi hai aur fir usi basis pe hamare evidence hame kitna support karte hai

Plain understanding

Bayes theorem = **One's gut feeling + logical support**😎